You'll use CloudFormation to describe the VPC with its subnets. The template is split into smaller parts to make it easier to read in the book. As usual, you'll find the code in the book's code repository on GitHub: https://github.com/AWSinAction/code3. The template is located at /chapter05/vpc.yaml.

5.5.1 Creating the VPC and an internet gateway (IGW)

The first resources listed in the template are the VPC and the internet gateway (IGW). In the next code snippet, the IGW will translate the public IP addresses of your virtual machines to their private IP addresses using network address translation (NAT). All public IP addresses used in the VPC are controlled by this IGW:

```
VPC:
  Type: 'AWS::EC2::VPC'
  Properties:
    CidrBlock: '10.0.0.0/16'                    ①
    EnableDnsHostnames: 'true'
    Tags:                                       ②
    - Key: Name
      Value: 'AWS in Action: chapter 5 (VPC)'
InternetGateway:                                ③
  Type: 'AWS::EC2::InternetGateway'
  Properties: {}
VPCGatewayAttachment:                           ④
  Type: 'AWS::EC2::VPCGatewayAttachment'
  Properties:
    VpcId: !Ref VPC
    InternetGatewayId: !Ref InternetGateway
```

① The IP address space used for the private network

② Adds a Name tag to the VPC

③ An IGW is needed to enable traffic to and from the internet.

④ Attaches the internet gateway to the VPC